FURTHER MECHANICS / CENTRE OF MASS

# Centre of Mass of Systems of Particles

Worked solutions 9 questions · 59 marks

EXAM BOARDS Edexcel FM2 9FM0/4C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Minor Y431 OCR MEI Mech Major Y421 · CIE 9231 Paper 3

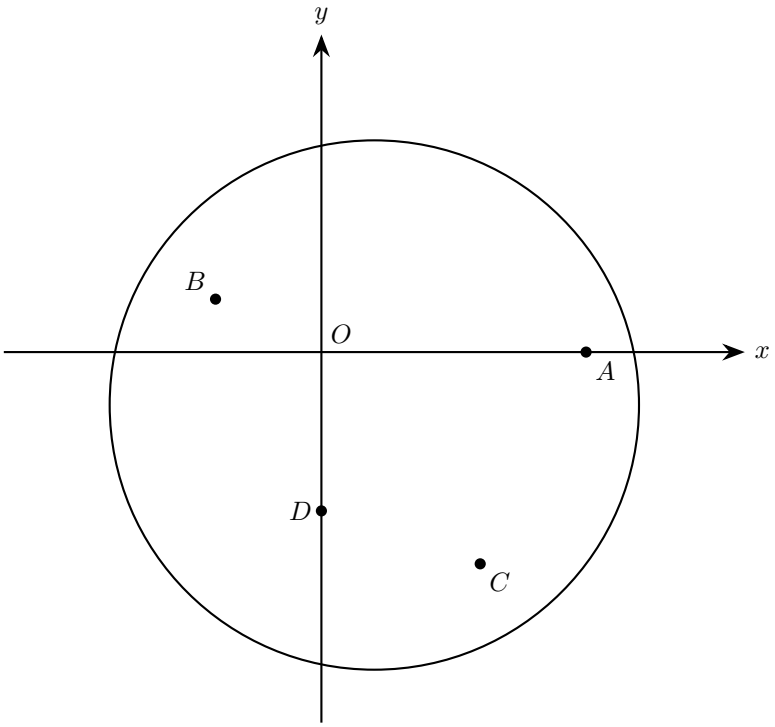
**Instructions**

Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question gives a value. Where a question leaves $g$ unspecified and a final answer depends on it, the solutions also give that answer for $g = 10\,\mathrm{m\,s^{-2}}$.

CIE (9231 Paper 3): papers take $g = 10\,\mathrm{m\,s^{-2}}$; use the $g = 10$ final answers where the solutions give them.

## Questions


| | | |
|---|---|---|
| Question 1 | *4 marks* | 2 |
| Question 2 | *5 marks* | 4 |
| Question 3 | *4 marks* | 6 |
| Question 4 | *4 marks* | 8 |
| Question 5 | *6 marks* | 9 |
| Question 6 | *9 marks* | 11 |
| Question 7 | *8 marks* | 13 |
| Question 8 | *7 marks* | 15 |
| Question 9 | *12 marks* | 17 |




**1.** Four sensors are attached to a circular plate.

The plate is to be modelled as a uniform lamina and the four sensors as four particles.

The diagram shows the lamina, the four particles $A$, $B$, $C$ and $D$, and the coordinate axes.

The lamina has mass 6 kg and its centre of mass is at the point $(1, -1)$.

Particle $A$ has mass 4 kg and is at the point $(5, 0)$.
Particle $B$ has mass 3 kg and is at the point $(-2, 1)$.
Particle $C$ has mass 5 kg and is at the point $(3, -4)$.
Particle $D$ has mass 2 kg and is at the point $(0, -3)$.

Find the coordinates of the centre of mass of the system of plate and sensors. **[4]**

**Solution**

Treat the uniform lamina as a particle of mass 6 kg acting at its centre of mass $(1, -1)$.

Let the centre of mass of the whole system be $(\bar{x}, \bar{y})$.

For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 6 + 4 + 3 + 5 + 2 = 20\,\text{kg}$$

For the $x$-coordinate:

$$\begin{aligned} 20\bar{x} &= 6(1) + 4(5) + 3(-2) + 5(3) + 2(0) \\ &= 6 + 20 - 6 + 15 + 0 \\ &= 35 \end{aligned}$$

Hence

$$\bar{x} = \frac{35}{20} = \frac{7}{4}$$

For the $y$-coordinate:

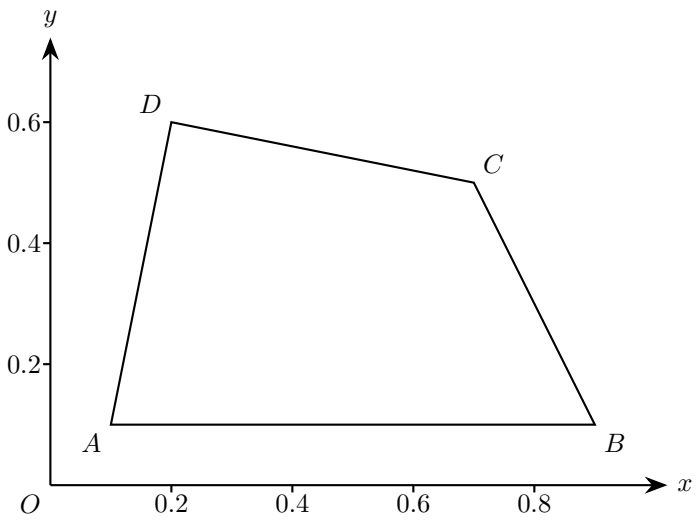
$$\begin{aligned} 20\bar{y} &= 6(-1) + 4(0) + 3(1) + 5(-4) + 2(-3) \\ &= -6 + 0 + 3 - 20 - 6 \\ &= -29 \end{aligned}$$

Hence

$$\bar{y} = -\frac{29}{20}$$

Therefore the centre of mass of the system is

$$\left(\frac{7}{4}, -\frac{29}{20}\right)$$




**2.** Particles $A$, $B$, $C$ and $D$ of masses 1 kg, 2 kg, 3 kg and 4 kg respectively are joined by light rigid rods to form the quadrilateral frame shown in the diagram. The particles are at the points $A(0.1, 0.1)$, $B(0.9, 0.1)$, $C(0.7, 0.5)$ and $D(0.2, 0.6)$, where coordinates are measured in metres.

$G$, the centre of mass of the frame, has coordinates $(\bar{x}, \bar{y})$.

**(a)** Show that $\bar{y} = 0.42$. Find $\bar{x}$. **[4]**

**(b)** Using your answer to part (a), explain why the frame cannot be in equilibrium in a horizontal plane if it is supported by a single peg touching the frame at one point. **[1]**

**Solution**

**(a)** For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 1 + 2 + 3 + 4 = 10\,\text{kg}$$

For the $y$-coordinate:

$$\begin{aligned} 10\bar{y} &= 1(0.1) + 2(0.1) + 3(0.5) + 4(0.6) \\ &= 0.1 + 0.2 + 1.5 + 2.4 = 4.2 \\ \bar{y} &= \frac{4.2}{10} = 0.42 \end{aligned}$$

So $\bar{y} = 0.42$, as required.

For the $x$-coordinate:

$$\begin{aligned} 10\bar{x} &= 1(0.1) + 2(0.9) + 3(0.7) + 4(0.2) \\ &= 0.1 + 1.8 + 2.1 + 0.8 = 4.8 \\ \bar{x} &= \frac{4.8}{10} = 0.48 \end{aligned}$$

Hence the centre of mass is

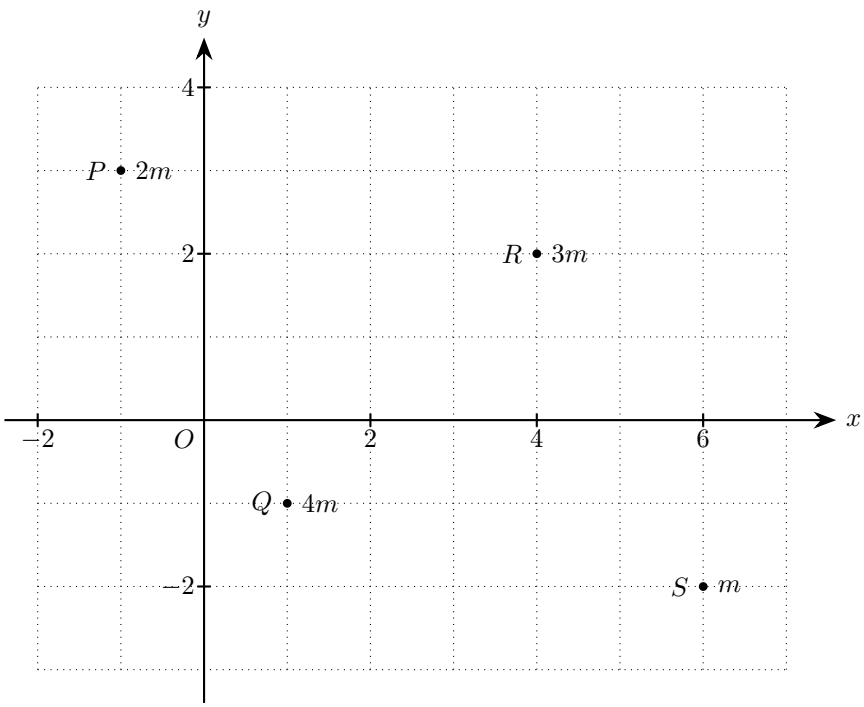
$$G = (\bar{x}, \bar{y}) = (0.48, 0.42)$$

**(b)** For equilibrium on a single peg, the peg must be vertically below the centre of mass.

So in the horizontal plane, the peg would have to touch the frame at $G = (0.48, 0.42)$.

But this point is not on the frame itself; it lies inside the quadrilateral, not on any of its sides.

Therefore the frame cannot be in equilibrium if it is supported by a single peg touching the frame at one point.




**3.** The diagram shows a system of four particles of masses $2m$, $4m$, $3m$ and $m$ situated in the $x$-$y$ plane at the points $P(-1, 3)$, $Q(1, -1)$, $R(4, 2)$ and $S(6, -2)$ respectively.

Find the coordinates of the centre of mass of the system. **[4]**

**Solution**

Let the centre of mass be $(\bar{x}, \bar{y})$.

For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 2m + 4m + 3m + m = 10m$$

For the $x$-coordinate:

$$M\bar{x} = 2m(-1) + 4m(1) + 3m(4) + m(6)$$

So

$$10m\bar{x} = -2m + 4m + 12m + 6m = 20m$$

Hence

$$\bar{x} = 2$$

For the $y$-coordinate:

$$M\bar{y} = 2m(3) + 4m(-1) + 3m(2) + m(-2)$$

So

$$10m\bar{y} = 6m - 4m + 6m - 2m = 6m$$

Hence

$$\bar{y} = \frac{3}{5}$$

Therefore the coordinates of the centre of mass are $(2, \frac{3}{5})$.

**4.** Four particles of mass $m$, $2m$, $4m$ and $\lambda m$ are placed at points with coordinates $(0,0)$, $(2,3)$, $(1,-1)$ and $(4,2)$ respectively. The centre of mass of the system is at $(2,k)$.

**(a)** Show that $\lambda = 3$. **[2]**

**(b)** Calculate the value of $k$. **[2]**

**Solution**

**(a)** For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = m + 2m + 4m + \lambda m = (7+\lambda)m$$

For the $x$-coordinate, with $\bar{x} = 2$:

$$M(2) = m(0) + 2m(2) + 4m(1) + \lambda m(4)$$

So

$$\begin{aligned} 2(7+\lambda)m &= 0 + 4m + 4m + 4\lambda m \\ 2(7+\lambda) &= 8 + 4\lambda \\ 14 + 2\lambda &= 8 + 4\lambda \\ 6 &= 2\lambda \\ \lambda &= 3 \end{aligned}$$

So $\lambda = 3$, as required.

**(b)** With $\lambda = 3$, the total mass is

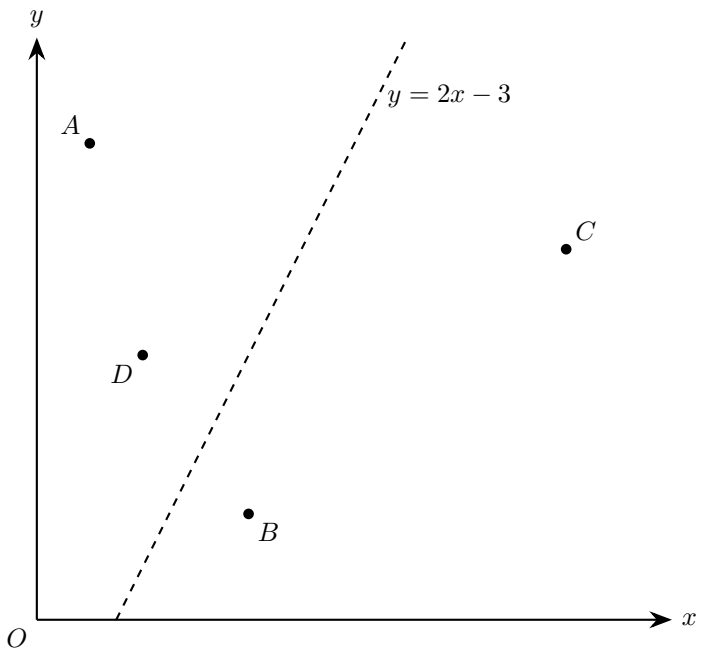
$$M = (7+3)m = 10m$$

For the $y$-coordinate, with $\bar{y} = k$:

$$Mk = m(0) + 2m(3) + 4m(-1) + 3m(2)$$

So

$$\begin{aligned} 10mk &= 0 + 6m - 4m + 6m \\ 10mk &= 8m \\ k &= \frac{8}{10} \\ k &= \frac{4}{5} \end{aligned}$$




**5.** The diagram shows four particles, $A$, $B$, $C$ and $D$, fixed in a horizontal plane which contains the $x$- and $y$-axes.

Particle $A$ has mass 3 kg and is attached at the point $(1, 9)$.
Particle $B$ has mass 4 kg and is attached at the point $(4, 2)$.
Particle $C$ has mass 5 kg and is attached at the point $(10, 7)$.
Particle $D$ has mass $m$ kg and is attached at the point $(2, 5)$.

The centre of mass of the four particles lies on the line $y = 2x - 3$.
Find the value of $m$ and the coordinates of the centre of mass. **[6]**

**Solution**

Let the centre of mass be $(\bar{x}, \bar{y})$.

For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 3 + 4 + 5 + m = (12 + m)\,\text{kg}$$

For the $x$-coordinate:

$$M\bar{x} = 3(1) + 4(4) + 5(10) + m(2)$$

so

$$\begin{aligned}(12 + m)\bar{x} &= 3 + 16 + 50 + 2m \\ &= 69 + 2m\end{aligned}$$

Hence

$$\bar{x} = \frac{69 + 2m}{12 + m}$$

For the $y$-coordinate:

$$M\bar{y} = 3(9) + 4(2) + 5(7) + m(5)$$

so

$$\begin{aligned}(12+m)\bar{y} &= 27 + 8 + 35 + 5m \\ &= 70 + 5m\end{aligned}$$

Hence

$$\bar{y} = \frac{70+5m}{12+m}$$

The centre of mass lies on the line $y = 2x - 3$, so

$$\frac{70+5m}{12+m} = 2\left(\frac{69+2m}{12+m}\right) - 3$$

Multiply through by $12 + m$:

$$\begin{aligned}70 + 5m &= 2(69 + 2m) - 3(12 + m) \\ &= 138 + 4m - 36 - 3m \\ &= 102 + m\end{aligned}$$

Therefore

$$\begin{aligned}70 + 5m &= 102 + m \\ 4m &= 32 \\ m &= 8\end{aligned}$$

Now substitute $m = 8$. Then the total mass is

$$M = 12 + 8 = 20\,\text{kg}$$

So

$$\begin{aligned}\bar{x} &= \frac{69 + 2(8)}{20} = \frac{85}{20} = \frac{17}{4} \\ \bar{y} &= \frac{70 + 5(8)}{20} = \frac{110}{20} = \frac{11}{2}\end{aligned}$$

Therefore

$$m = 8$$

and the centre of mass is

$$\left(\frac{17}{4}, \frac{11}{2}\right)$$

**6.** Four particles of masses $m$, $2m$, $3m$ and $km$ are positioned at the points with coordinates $(-a, a)$, $(2a, -2a)$, $(0, 0)$ and $(\mu a, 2\mu a)$ respectively, where $k$ and $\mu$ are constants.

The centre of mass of the four particles is at the point with coordinates $(2a, 3a)$.

**(a)** Find the value of $k$. **[6]**

**(b)** Find the value of $\mu$. **[3]**

**Solution**

**(a)** For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = m + 2m + 3m + km = (6 + k)m$$

For the $x$-coordinate, with $\bar{x} = 2a$:

$$M(2a) = m(-a) + 2m(2a) + 3m(0) + km(\mu a)$$

so

$$\begin{aligned}(6 + k)m(2a) &= -am + 4am + 0 + k\mu am \\ 2(6 + k) &= 3 + k\mu \\ 12 + 2k &= 3 + k\mu \\ k\mu &= 9 + 2k\end{aligned}$$

For the $y$-coordinate, with $\bar{y} = 3a$:

$$M(3a) = m(a) + 2m(-2a) + 3m(0) + km(2\mu a)$$

so

$$\begin{aligned}(6 + k)m(3a) &= am - 4am + 0 + 2k\mu am \\ 3(6 + k) &= -3 + 2k\mu \\ 18 + 3k &= -3 + 2k\mu \\ 2k\mu &= 21 + 3k\end{aligned}$$

Now double the equation $k\mu = 9 + 2k$:

$$2k\mu = 18 + 4k$$

Equating the two expressions for $2k\mu$:

$$\begin{aligned}18 + 4k &= 21 + 3k \\ k &= 3\end{aligned}$$

**(b)** Substitute $k = 3$ into

$$k\mu = 9 + 2k$$

to get

$$3\mu = 9 + 2(3)$$

$$3\mu = 15$$

$$\mu = 5$$

**7.** Three particles of masses $m$, $2m$ and $km$ are placed at the points $A(a, 0)$, $B(-2a, 3a)$ and $C(2a, -a)$ respectively, where $k > 0$.

The centre of mass of the three particles is the point with coordinates $(\bar{x}, \bar{y})$.

**(a)** (i) Find $\bar{x}$ in terms of $a$ and $k$.
(ii) Find $\bar{y}$ in terms of $a$ and $k$. **[4]**

**(b)** Given that the centre of mass lies on the circle with centre $(a, 0)$ and radius $\frac{\sqrt{2}}{2}a$, find the possible values of $k$. **[4]**

**Solution**

**(a)** For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = m + 2m + km = (k + 3)m$$

**(i)** For the $x$-coordinate:

$$M\bar{x} = m(a) + 2m(-2a) + km(2a)$$

So

$$\begin{aligned}(k + 3)m\bar{x} &= am - 4am + 2kam \\ &= am(2k - 3)\end{aligned}$$

Hence

$$\bar{x} = \frac{a(2k - 3)}{k + 3}$$

**(ii)** For the $y$-coordinate:

$$M\bar{y} = m(0) + 2m(3a) + km(-a)$$

So

$$\begin{aligned}(k + 3)m\bar{y} &= 0 + 6am - kam \\ &= am(6 - k)\end{aligned}$$

Hence

$$\bar{y} = \frac{a(6 - k)}{k + 3}$$

**(b)** The centre of mass lies on the circle with centre $(a, 0)$ and radius $\frac{\sqrt{2}}{2}a$, so

$$(\bar{x} - a)^2 + \bar{y}^2 = \left(\frac{\sqrt{2}}{2}a\right)^2 = \frac{a^2}{2}$$

Using part (a):

$$\begin{aligned}\bar{x} - a &= \frac{a(2k-3)}{k+3} - a \\ &= a\left(\frac{2k-3-(k+3)}{k+3}\right) \\ &= \frac{a(k-6)}{k+3}\end{aligned}$$

and

$$\bar{y} = \frac{a(6-k)}{k+3} = -\frac{a(k-6)}{k+3}$$

Substituting into the circle equation:

$$\begin{aligned}\left(\frac{a(k-6)}{k+3}\right)^2 + \left(-\frac{a(k-6)}{k+3}\right)^2 &= \frac{a^2}{2} \\ 2\left(\frac{a(k-6)}{k+3}\right)^2 &= \frac{a^2}{2}\end{aligned}$$

So

$$4(k-6)^2 = (k+3)^2$$

Expanding:

$$\begin{aligned}4(k^2 - 12k + 36) &= k^2 + 6k + 9 \\ 4k^2 - 48k + 144 &= k^2 + 6k + 9 \\ 3k^2 - 54k + 135 &= 0 \\ k^2 - 18k + 45 &= 0\end{aligned}$$

Factorising:

$$(k-3)(k-15) = 0$$

Hence the possible values of $k$ are

$$k = 3 \quad \text{or} \quad k = 15$$

**8.** Three particles of masses $2m$, $5m$ and $3m$ are placed at the points $(1, -4)$, $(-3, 2)$ and $(q, 2q - 1)$ respectively.

The value of $q$ is such that the distance of the centre of mass of the three particles from the point $(1, 0)$ is as small as possible.

Find the value of $q$. **[7]**

**Solution**

Let the centre of mass be $G(\bar{x}, \bar{y})$.

For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 2m + 5m + 3m = 10m$$

For the $x$-coordinate:

$$10m\bar{x} = 2m(1) + 5m(-3) + 3m(q)$$

so

$$10\bar{x} = 2 - 15 + 3q$$

$$\bar{x} = \frac{3q - 13}{10}$$

For the $y$-coordinate:

$$10m\bar{y} = 2m(-4) + 5m(2) + 3m(2q - 1)$$

so

$$10\bar{y} = -8 + 10 + 6q - 3$$

$$= 6q - 1$$

$$\bar{y} = \frac{6q - 1}{10}$$

Hence

$$G\left(\frac{3q - 13}{10}, \frac{6q - 1}{10}\right)$$

Let the distance from $G$ to $(1, 0)$ be $d$. Then

$$d^2 = \left(\frac{3q - 13}{10} - 1\right)^2 + \left(\frac{6q - 1}{10}\right)^2$$

$$= \left(\frac{3q - 23}{10}\right)^2 + \left(\frac{6q - 1}{10}\right)^2$$

$$= \frac{(3q - 23)^2 + (6q - 1)^2}{100}$$

Now expand:

$$(3q-23)^2+(6q-1)^2 = 9q^2-138q+529+36q^2-12q+1$$
$$= 45q^2-150q+530$$

So

$$d^2 = \frac{45q^2-150q+530}{100}$$

To make $d$ as small as possible, it is enough to make $d^2$ as small as possible. Complete the square:

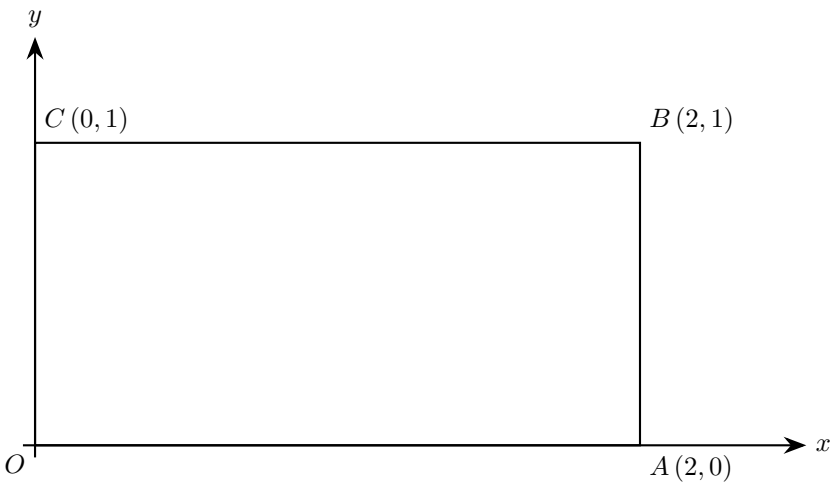
$$45q^2-150q+530 = 45\left(q^2-\frac{10}{3}q\right)+530$$
$$= 45\left[\left(q-\frac{5}{3}\right)^2-\frac{25}{9}\right]+530$$
$$= 45\left(q-\frac{5}{3}\right)^2-125+530$$
$$= 45\left(q-\frac{5}{3}\right)^2+405$$

Since $45\left(q-\frac{5}{3}\right)^2 \geq 0$, this expression is smallest when

$$q-\frac{5}{3}=0$$

Therefore

$$q=\frac{5}{3}$$




**9.** The diagram shows a light rectangular lamina $OABC$ where $O$ is the origin of the coordinate system in which the points $A$, $B$ and $C$ have coordinates $(2, 0)$, $(2, 1)$ and $(0, 1)$ respectively. Coordinates refer to the axes shown and the units are metres.

The following three forces are applied to the lamina.

$\begin{pmatrix} p \\ 2q \end{pmatrix}$ N at the point $(1, 0)$

$\begin{pmatrix} -q \\ p \end{pmatrix}$ N at the point $C$

$\begin{pmatrix} -3 \\ -6 \end{pmatrix}$ N at the point $(t, 1)$, where $0 \leq t \leq 2$

The three forces form a couple of magnitude 3 N m in the anticlockwise direction.

**(a)** Determine the values of $p$, $q$ and $t$. **[5]**

The three forces are removed. The lamina is in the same position as shown in the diagram.

Three particles are now attached to certain points on the lamina as follows.

**(b)** Find the coordinates of the composite body's centre of mass. Give your answer in terms of $\lambda$, where the particles are

- a particle of mass 2 kg at the point $(2\lambda^2, \lambda)$, where $0 \leq \lambda \leq 1$
- a particle of mass 1 kg at the point $A$
- a particle of mass 1 kg at the point $C$

**[4]**

**(c)** Show that, as $\lambda$ varies, the centre of mass of the combined object lies on the curve

$$x = 4y^2 - 2y + k$$

where $k$ is a constant to be determined. **[3]**

**Solution**

**(a)** Since the three forces form a couple, their resultant force is zero.

Resolving in the $x$- and $y$-directions:

$$p - q - 3 = 0, \qquad p + 2q - 6 = 0$$

Subtracting the first equation from the second gives

$$3q - 3 = 0$$

so

$$q = 1$$

Substituting into $p - q - 3 = 0$:

$$p - 1 - 3 = 0 \quad \Longrightarrow \quad p = 4$$

Now take moments about $O$, with anticlockwise positive, treating the two components of each force separately.

For the force at $(1, 0)$, the $x$-component acts along the $x$-axis, through $O$. The $y$-component, $2q$ N in the positive $y$-direction, acts at a perpendicular distance of 1 m from $O$, so its moment is anticlockwise.

For the force at $C$, the $y$-component acts along the $y$-axis, through $O$. The $x$-component, $q$ N in the negative $x$-direction, acts at a perpendicular distance of 1 m from $O$, so its moment is anticlockwise.

For the force at $(t, 1)$, the $x$-component, 3 N in the negative $x$-direction, acts at a perpendicular distance of 1 m from $O$, so its moment is anticlockwise. The $y$-component, 6 N in the negative $y$-direction, acts at a perpendicular distance of $t$ m from $O$, so its moment is clockwise.

The forces form an anticlockwise couple of magnitude 3 N m, so

$$2q(1) + q(1) + 3(1) - 6t = 3$$

Using $q = 1$:

$$2 + 1 + 3 - 6t = 3$$

Therefore

$$6 - 6t = 3 \quad \Longrightarrow \quad 6t = 3 \quad \Longrightarrow \quad t = \frac{1}{2}$$

So

$$p = 4, \qquad q = 1, \qquad t = \frac{1}{2}$$

**(b)** The lamina is light, so its mass is taken to be zero.

Let the centre of mass of the composite body be $(\bar{x}, \bar{y})$.

For particles of masses $m_i$ at the points $(x_i, y_i)$, the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 2 + 1 + 1 = 4 \text{ kg}$$

For the $x$-coordinate:

$$4\bar{x} = 2(2\lambda^2) + 1(2) + 1(0) = 4\lambda^2 + 2$$

So

$$\bar{x} = \lambda^2 + \frac{1}{2}$$

For the $y$-coordinate:

$$4\bar{y} = 2(\lambda) + 1(0) + 1(1) = 2\lambda + 1$$

So

$$\bar{y} = \frac{2\lambda + 1}{4}$$

Hence the centre of mass is

$$\left(\lambda^2 + \frac{1}{2}, \frac{2\lambda + 1}{4}\right)$$

**(c)** From part (b), the centre of mass has coordinates

$$\bar{x} = \lambda^2 + \frac{1}{2}, \qquad \bar{y} = \frac{2\lambda + 1}{4}$$

Rearranging the expression for $\bar{y}$:

$$4\bar{y} = 2\lambda + 1$$

so

$$\lambda = \frac{4\bar{y} - 1}{2}$$

Substitute this into the expression for $\bar{x}$:

$$\begin{aligned}
\bar{x} &= \left(\frac{4\bar{y} - 1}{2}\right)^2 + \frac{1}{2} \\
&= \frac{16\bar{y}^2 - 8\bar{y} + 1}{4} + \frac{1}{2} \\
&= 4\bar{y}^2 - 2\bar{y} + \frac{1}{4} + \frac{1}{2} \\
&= 4\bar{y}^2 - 2\bar{y} + \frac{3}{4}
\end{aligned}$$

Therefore, as $\lambda$ varies, the centre of mass lies on the curve

$$x = 4y^2 - 2y + k$$

with

$$k = \frac{3}{4}$$